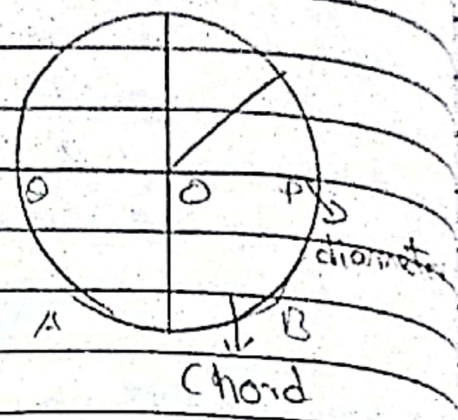
13/10/25

Circles

~~Circle - It is~~

Recap:

The Diameter of circle = OP
Chord = AB
Radius = OP



Introduction

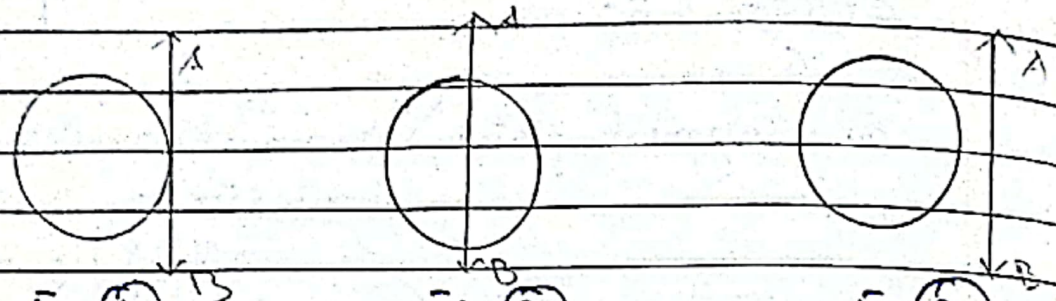


Fig (1) Fig (2) Fig (3)

(1) Fig (1) the line segment AB is the tangent.

(2) Fig (2) the line segment AB is a secant.

(3) Fig (3) AB not intersecting lines.

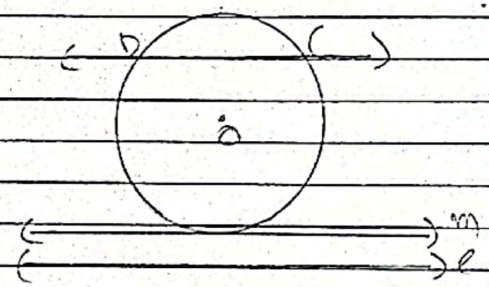
Ex-10.1

1-Ans (d) $\sqrt{119}$ cm

Reason: we have $OP = 5$ cm
and $OQ = 12$ cm
Since, PQ is the tangent and OP is the radius,
$\therefore \angle OPQ = 90^\circ$
$\Rightarrow \Delta OPQ$ is right angled triangle.

By Pythagoras' Theorem,
$OQ^2 = OP^2 + PQ^2$
$(12)^2 = (5)^2 + PQ^2$
$144 = 25 + PQ^2$
$PQ^2 = 119$
Hence, $PQ = \sqrt{119}$ cm

4-Solu Let l be the given line.
Draw a circle with centre O.

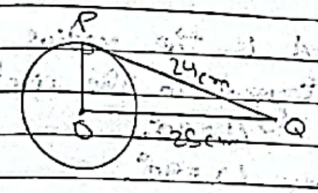


Let m and n be two lines parallel to line l such that m is a tangent as well as parallel to l and n is secant to the circle as well as parallel to l.

Ex-10.2

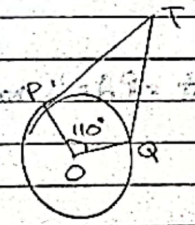
1-Ans (a) 7 cm

Justification: Here $OQ = 25$ cm, $PQ = 24$ cm.
$\therefore \angle QPO = 90^\circ$.



In right angled ΔOPQ,
$OQ^2 = PQ^2 + OP^2$
$\Rightarrow (25)^2 = OP^2 + (24)^2$
$\Rightarrow OP^2 = 625 - 576$
$\Rightarrow OP^2 = 49 = (7)^2$
Hence, $OP = 7$

2-Solu (b)
Justification: $\angle OPT = 90^\circ$
$\angle OQT = 90^\circ$
and $\angle POQ = 110^\circ$ (Given)
Since, POQT is a quadrilateral,
$\therefore \angle POQ + \angle OQT + \angle PTQ + \angle TPO = 360^\circ$
$\Rightarrow 110^\circ + 90^\circ + \angle PTQ + 90^\circ = 360^\circ$.
Hence, $\angle PTQ = 360^\circ - 290^\circ = 70^\circ$



4-Solu Given: A circle with centre O.

AB is the diameter of the circle. l and m are tangents at A and B respectively.

To prove: $l \parallel m$

Proof: Since OA is the radius and 'l' is the tangent to the circle,

$\therefore \angle 1 = 90°$.

Similarly, $\angle 2 = 90°$.

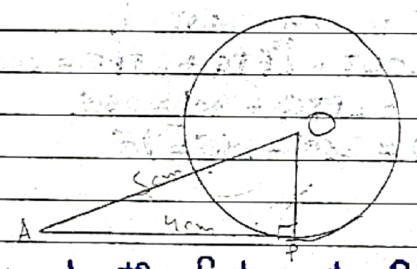
Thus, $\angle 1 = \angle 2 = 90°$.

But these form a pair of alternate angles.

$\therefore l \parallel m$

Hence, the tangents drawn at the ends of a diameter of a circle are parallel.

6-Solu Given a circle with centre 'O'. A is any point outside the circle at a distance of 5cm from the centre O.



Now, length of tangent = PA = 4cm.

Since, the radius OP is perpendicular to the tangent at P,

$\therefore \angle OPA = 90°$

In right ΔOPA,

$OA^2 = OP^2 + PA^2$ [Pythagoras Theorem]

$\Rightarrow (5)^2 = OP^2 + (4)^2$

$\Rightarrow OP^2 = 25 - 16 = 9 = (3)^2$

$\Rightarrow OP = 3$ cm

Hence, the radius of the circle is 3 cm.

8-Solu Given: A quadrilateral ABCD which circumscribes a circle with centre O.

To prove: $AB + CD = AD + BC$.

Proof: Since the lengths of tangents drawn from an external point are equal,

$\therefore BQ = BP$ — (1)

Similarly, $AS = AP$ — (2)

$CQ = CR$ — (3)

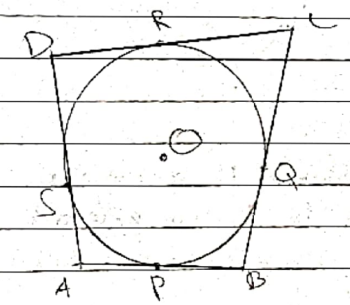
and $DS = DR$ — (4)

Adding (1), (2), (3) and (4), we get:

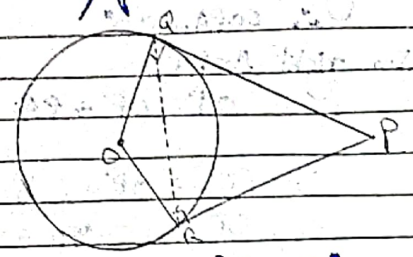
$(BQ + CQ) + (AS + DS) = (BP + AP) + (CR + DR)$

$\Rightarrow BC + AD = AB + CD$

Hence, $AB + CD = AD + BC$.



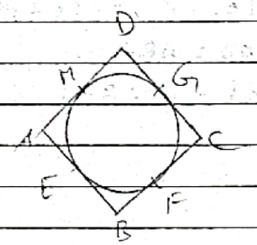
10-Soln Given: A circle with centre O.
P is any point outside the circle.



To prove: $\angle ROQ + \angle QPR = 180^\circ$

Proof: Since OQ is the radius and PQ is the tangent from P to the circle;
$\therefore \angle OQP = 90^\circ$ — (1)
Similarly, $\angle ORP = 90^\circ$ — (2)
In quadrilateral ROQP,
$\angle ROQ + \angle PRO + \angle OQP + \angle QPR = 360^\circ$
$\Rightarrow \angle ROQ + 90^\circ + 90^\circ + \angle QPR = 360^\circ$
(using (1) & (2))
$\Rightarrow \angle ROQ + \angle QPR = 360^\circ - 180^\circ$
$\Rightarrow$ Hence, $\angle ROQ + \angle QPR = 180^\circ$

11-Soln Given: A parallelogram ABCD circumscribing the circle with centre O.



To prove: ABCD is a rhombus.
Proof: Here B is an external point,

$\therefore$ BE = BF — (1)
Similarly, AE = AH — (2)
CG = CF — (3)
and DG = DH — (4)
Adding (1), (2), (3) and (4), we get:
$(BE + AE) + (CG + DG) = (BF + CF) + (AH + DH)$
$\Rightarrow AB + CD = BC + AD$ — (5)
Since, ABCD is a parallelogram,
$\therefore AB = CD$ and $BC = AD$ — (6)
From (5) and (6),
$AB + AB = BC + BC$
$\Rightarrow 2AB = 2BC$
$\Rightarrow AB = BC$
Thus, $AB = BC = CD = AD$
Hence, ABCD is a rhombus.

12-Soln Given: A quadrilateral ABCD circumscribing a circle having centre O.
The sides AB, BC, CD, DA touch the circle at H, E, F, G respectively.

To prove: $\angle AOB + \angle DOC = 180^\circ$ and
$\angle DOA + \angle COB = 180^\circ$

Construction: Join OA, OH, OB, OE, OC, OF, OD, OG

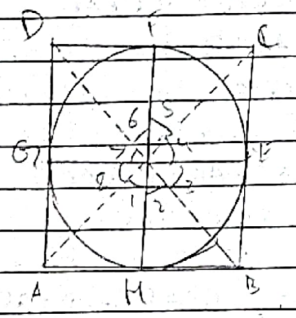
Proof: Since, the two tangents drawn from an external point subtend equal angles at the centre.
$\therefore \angle 2 = \angle 3$; $\angle 4 = \angle 5$; $\angle 6 = \angle 7$ and $\angle 8 = \angle 1$ — (1)
Since the sum of all $\angle$s at a point = 360°,
$\therefore \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 = 360^\circ$
$\Rightarrow \angle 1 + \angle 2 + \angle 2 + \angle 5 + \angle 5 + \angle 6 + \angle 6 + \angle 1 = 360^\circ$
$\Rightarrow 2(\angle 1 + \angle 2 + \angle 5 + \angle 6) = 360^\circ$

$\angle 1 + \angle 2 + \angle 5 + \angle 6 = 180^\circ$

$\angle AOB + \angle DOC = 180^\circ$

Similarly, $\angle DOA + \angle COB = 180^\circ$

Hence, opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.



W.S.